

SUPUNG, Korea, Democratic People's Republic of

WMO: 470280

Lat: 40.450N

Lon: 124.933E

Elev: 76

StdP: 100.42

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.2	-15.4	-23.5	0.5	-13.7	-21.6	0.5	-13.0	6.2	-9.6	5.5	-9.2	1.6	180	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	31.4	24.2	30.0	23.4	28.8	22.9	26.1	29.5	25.1	28.0	24.3	27.0	1.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.4	28.4	24.2	19.3	26.9	23.5	18.5	26.0	81.8	29.7	77.5	28.0	74.2	27.1	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.4	3.6	DB	-20.0	34.4	2.4	1.5	-21.7	35.5	-23.1	36.4	-24.4	37.2	-26.2	38.3	(4)
(5)			WB	-20.1	27.4	2.5	1.4	-21.9	28.4	-23.4	29.2	-24.8	29.9	-26.6	31.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-7.3	-4.4	2.4	10.0	16.4	21.3	24.5	24.1	18.9	11.7	3.1	-5.2	(6)	
	DBStd	11.69	3.87	4.07	3.77	3.76	2.86	2.07	1.89	2.28	2.83	3.56	4.75	4.36	(7)	
	HDD10.0	1926	535	403	237	46	1	0	0	0	0	23	212	469	(8)	
	HDD18.3	3669	793	636	494	251	73	3	0	1	27	206	458	728	(9)	
	CDD10.0	1815	0	0	1	44	198	338	449	438	266	77	4	0	(10)	
	CDD18.3	517	0	0	0	0	13	91	191	180	43	1	0	0	(11)	
	CDH23.3	3523	0	0	0	8	179	607	1282	1225	219	3	0	0	(12)	
	CDH26.7	916	0	0	0	0	25	142	372	356	22	0	0	0	(13)	
(14) Wind	WSAvg	1.2	1.4	1.2	1.2	1.2	1.1	0.9	0.8	0.9	1.1	1.4	1.5	1.5	(14)	
(15) Precipitation	PrecAvg	1066	8	18	22	56	80	125	294	256	91	57	43	17	(15)	
	PrecMax	1795	29	71	71	99	191	248	683	784	300	163	99	53	(16)	
	PrecMin	697	0	1	1	1	16	23	133	37	14	2	7	2	(17)	
	PrecStd	253	7	18	15	26	40	58	125	154	61	37	29	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	8.1	16.8	24.5	29.4	31.6	33.9	33.4	28.7	23.3	16.0	7.2	(19)	
		MCWB	1.8	4.0	8.7	13.6	18.7	21.0	25.8	26.6	22.0	16.8	11.9	5.1	(20)	
	2%	DB	2.3	5.5	13.5	22.1	27.1	29.8	31.7	31.2	27.0	21.0	13.9	4.7	(21)	
		MCWB	0.3	1.7	7.5	13.2	17.5	20.7	24.7	25.1	20.7	15.1	10.2	2.8	(22)	
	5%	DB	0.7	3.7	11.1	19.8	25.4	28.3	30.1	29.9	25.8	19.4	11.9	2.8	(23)	
		MCWB	-1.1	1.0	6.0	11.6	16.6	20.4	24.1	24.1	19.4	14.2	8.8	1.1	(24)	
	10%	DB	-1.0	2.2	9.1	17.6	23.7	26.9	28.7	28.6	24.5	17.7	10.1	1.2	(25)	
		MCWB	-2.8	-0.4	4.7	10.5	15.9	20.1	23.8	23.6	18.8	13.2	7.5	-0.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	4.6	10.3	16.0	20.7	23.5	28.0	27.6	23.4	17.7	12.7	5.9	(27)	
		MCDB	3.6	7.1	14.1	20.8	26.9	28.4	32.4	32.0	27.2	21.5	14.9	7.0	(28)	
	2%	WB	0.7	2.6	8.3	14.1	19.0	22.5	26.4	26.2	22.0	16.2	10.9	3.1	(29)	
		MCDB	2.1	4.6	12.6	19.8	24.6	27.0	29.7	29.8	25.6	19.8	13.3	4.4	(30)	
	5%	WB	-1.1	1.2	6.6	12.7	17.9	21.7	25.2	25.3	20.9	15.1	9.4	1.3	(31)	
		MCDB	0.5	3.3	10.3	18.5	23.5	26.1	28.1	28.4	24.2	18.0	11.5	2.5	(32)	
	10%	WB	-2.7	-0.2	5.1	11.1	16.8	21.0	24.5	24.6	20.0	14.1	7.6	-0.2	(33)	
		MCDB	-1.1	2.0	8.7	16.4	21.9	25.2	27.2	27.2	22.9	16.9	9.7	1.1	(34)	
(35) Mean Daily Temperature Range	MDBR		9.3	9.7	10.2	12.1	12.3	10.1	7.6	8.2	10.4	10.6	8.2	7.9	(35)	
	5% DB	MCDBR	9.8	11.0	13.5	15.7	15.8	13.5	10.5	10.2	11.7	12.1	10.4	8.2	(36)	
		MCWBR	8.0	8.1	8.3	7.9	7.2	5.7	4.5	4.5	5.5	6.8	7.3	6.9	(37)	
	5% WB	MCDBR	9.4	9.9	12.6	14.3	13.4	10.1	8.2	8.0	9.2	10.5	9.0	7.8	(38)	
		MCWBR	7.9	7.9	8.4	7.9	6.9	5.0	4.3	4.2	4.9	6.6	6.9	6.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.392	0.473	0.539	0.626	0.656	0.731	0.698	0.596	0.523	0.551	0.479	0.405	(40)	
	taud		1.944	1.792	1.719	1.609	1.591	1.524	1.633	1.834	1.938	1.774	1.863	1.946	(41)	
	Ebn at Noon		710	699	704	676	670	621	637	693	714	617	606	654	(42)	
	Edh at Noon		130	177	212	253	263	282	251	200	169	178	142	121	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.46	3.21	4.29	4.87	5.51	4.92	4.12	4.41	4.36	3.27	2.26	2.05	(44)	
	RadStd		0.12	0.20	0.30	0.35	0.42	0.57	0.47	0.53	0.36	0.21	0.25	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page